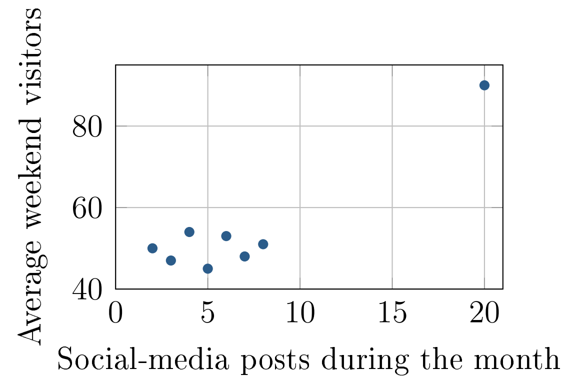
What Does This Correlation Support?

Unit 5 · Topic 5.2 · TODAY'S PROBLEM

Suppose a nonprofit records monthly social-media posts and average weekend visitors at eight community arts centers. The pairs are $(2, 50)$, $(3, 47)$, $(4, 54)$, $(5, 45)$, $(6, 53)$, $(7, 48)$, $(8, 51)$, and $(20, 90)$. Technology gives $r = 0.922$ for all eight centers and $r = 0.095$ for the first seven.



Omar: Use that strong correlation to forecast the attendance gained from additional posts; every observed center belongs in the analysis.

Tessa: Report the full analysis, but also examine the seven-center cluster and how posting levels were chosen before recommending a campaign.

FIRST TAKE (5 min, on your sheet)

Which proposal seems more defensible, Omar's or Tessa's? Cite one number, plot feature, or design fact.

DOK 1 **(a)** What do the sign and size of the full-data $r = 0.922$ tell you?

DOK 2 **(b)** Compare the two given correlations. Describe the plot feature that could explain the difference.

DOK 3 **(c)** Which approach would you recommend? Use both correlations and the plot, explaining why the full result should still be reported. Explain why observing posting levels does not show that extra posts cause more visitors. Write 3–4 sentences.

Optional review: follow-along (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.